

Namma-Santhe Ledger

Project #56 - Android App Development using Gen AI

■ Finance & Micro-Business Management for Village Markets

Empowering Weekly Santhe Vendors with Digital Bookkeeping

SUBMITTED BY	PROGRAMME	SEMESTER	INSTITUTE	COMPANY
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DOCUMENT TYPE	PLATFORM	DOMAIN	VERSION	
Product Design & Requirements	Android (Kotlin / Java)	FinTech / Rural Retail	v1.0 — 2025	
500M+	~40%	5 sec	Zero	
Rural Vendors in India	Sales on Credit (Udari)	Max Transaction Entry Time	Paper Diaries Needed	

1. Executive Summary & Problem Statement

Background

Weekly village markets — called **Santhe** — are the heartbeat of rural retail across Karnataka and much of India. Thousands of small vendors gather every week to sell vegetables, bangles, snacks, and daily essentials. These micro-entrepreneurs operate almost entirely on trust — and on **Udari** (informal credit), where regular customers buy today and pay at the next market.

The Problem

- **Memory Failure:** Vendors rely on pocket diaries or pure memory to track who owes them what. A busy market day can involve 50-100 transactions across dozens of credit customers.
- **Revenue Leakage:** Forgotten debts mean money never recovered. Studies suggest 15-25% of credit sales are never collected in unorganised retail.
- **No Daily Visibility:** A vendor has no quick way to know their *actual* net earnings after subtracting pending dues.

- **No Follow-up Mechanism:** There is no structured way to remind a customer about an outstanding balance before the next Santhe.
- **Paper Fragility:** Physical diaries get wet, torn, or lost — wiping out months of records.

Core Insight: The solution must be **faster than writing in a diary** — otherwise vendors will not adopt it. Every design decision flows from this single constraint: *entry in under 5 seconds*.

2. Product Vision & Value Proposition

Namma-Santhe Ledger is a **Simplified Digital Khata** designed exclusively for the smallest vendors at rural weekly markets. It replaces the messy pocket diary with a smartphone app that is as fast, simple, and reliable as jotting in a notebook — but infinitely smarter.

For Whom	Pain Point Solved	Value Delivered
Vegetable / bangle / snack vendors	Forgotten Udari debts	Instant credit ledger; zero memory load
Vendors with 20-100 regular customers	No daily profit visibility	One-tap Daily Summary screen
Semi-literate / low-digital-literacy users	Complex app interfaces	Large keypad, Kannada-ready labels
Vendors who extend repeat credit	No reminder mechanism	WhatsApp reminder in one tap

3. Scope of the Application

3.1 In Scope

- ✓ Customer management: add, edit, search customers by name.
- ✓ Credit (Udari) entry: record new credit transactions against a customer.
- ✓ Payment entry: record partial or full repayments from customers.
- ✓ Net balance calculation per customer and overall outstanding total.
- ✓ Daily Sales & Dues Summary: date-wise aggregated view.
- ✓ WhatsApp reminder: pre-filled message sent via WhatsApp Intent.
- ✓ Local data persistence using Room DB (offline-first).
- ✓ Basic analytics: total sold today, total pending dues, top debtors.
- ✓ Search / filter customer ledger.
- ✓ GenAI-powered smart insights: auto-suggest risky debtors, unusual patterns.

3.2 Out of Scope (v1.0) X Inventory management or stock tracking.

X GST / invoice generation.

- ✗ Multi-user / cloud sync across devices.
- ✗ Payment gateway integration (UPI, cards).
- ✗ Profit-loss accounting or balance sheet.
- ✗ Web or iOS versions.

3.3 Future Scope (v2.0+)

- Cloud backup via Firebase / Firestore.
- Voice entry in Kannada / regional languages using Gemini Speech API.
- SMS reminders for customers without WhatsApp.
- Group / family debt tracking.
- AI-driven credit-risk score per customer.
- Integration with ONDC or Sahay digital credit platforms.

4. App Usage & User Flow

The entire UX is designed around a single principle: **a vendor mid-transaction cannot spare more than 5 seconds**. Every flow is optimised for minimal taps.

Step	Action	Screen	Time Target
1	Open app — Home screen shows Total Outstanding prominently	Home Dashboard	0 s
2	Tap '+ Udari' FAB button	Quick Entry Sheet	1 s
3	Select customer (search or recent list)	Customer Picker	2 s
4	Enter amount on large numeric keypad	Amount Entry	3 s
5	Tap 'Add Udari' — transaction saved instantly	Confirmation Toast	5 s
—	To record payment: tap customer > 'Got Payment' > enter amount	Payment Log	<8 s
—	End of day: tap 'Summary' for Today Sold / Dues Pending	Daily Summary	Instant
—	Tap WhatsApp icon on customer row to send reminder	WhatsApp Intent	2 taps

5. Functional Requirements

FR-1: Customer Management

ID	Requirement
FR-1.1	Add new customer with name and optional phone number.
FR-1.2	Edit or archive existing customer records.
FR-1.3	Search customers by name (real-time, minimum 2 characters).
FR-1.4	Display each customer's current net balance (Dues / Advance).
FR-1.5	Sort customer list by balance (highest debt first by default).

FR-2: Transaction Entry

ID	Requirement
FR-2.1	Record a new Udari (credit) transaction in max 2 steps.
FR-2.2	Record a payment (partial or full) against a customer.
FR-2.3	Auto-timestamp each transaction; allow date override.
FR-2.4	Support optional note/item description per transaction.
FR-2.5	Prevent negative balance recording without confirmation.

FR-3: Ledger & Balance

ID	Requirement
FR-3.1	Show individual transaction history per customer (chronological).
FR-3.2	Calculate and display net balance per customer dynamically via ViewModel.
FR-3.3	Display Total Outstanding Dues on the Home Screen at all times.
FR-3.4	Support undo for the last transaction (within 30 seconds).

FR-4: Daily Summary

ID	Requirement
FR-4.1	Show 'Total Sold Today' (sum of all Udari entries for current date).
FR-4.2	Show 'Payments Received Today'.
FR-4.3	Show 'Net Cash Today' = Payments Received - New Udari Given.
FR-4.4	Show 'Top 5 Debtors' by outstanding amount.

FR-4.5	Date picker to view summary for any past market day.
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FR-5: WhatsApp Reminder

ID	Requirement
FR-5.1	Generate a pre-filled, friendly reminder message in Kannada/English.
FR-5.2	Open WhatsApp via Intent with the customer's phone number pre-filled.
FR-5.3	Message must include customer name, amount due, and vendor name.

FR-6: GenAI Features

ID	Requirement
FR-6.1	Smart Insight: flag customers who have not paid in 2+ market cycles.
FR-6.2	Auto-summarise vendor's financial health in plain language (e.g., 'You are owed 18% more than last week').
FR-6.3	Suggest optimal reminder timing based on past payment patterns.

6. Non-Functional Requirements

Category	Requirement	Target / Metric
Performance	Transaction entry to confirmation	< 500 ms
Performance	Home screen load time	< 1.5 s
Performance	Search results on >500 customers	< 300 ms
Reliability	App crash rate	< 0.1%
Reliability	Data persistence (Room DB) — zero data loss	100%
Offline	Full functionality without internet (except GenAI / WhatsApp)	Yes
Usability	New user (semi-literate) can complete first transaction	< 3 min
Usability	Minimum touch target size (WCAG AA)	48 x 48 dp
Accessibility	Support system font scaling up to 200%	Yes
Accessibility	TalkBack screen-reader compatibility	Yes

Security	All local data encrypted at rest (AES-256 via SQLCipher)	Yes
Security	Optional app lock via PIN / biometric	Yes
Compatibility	Minimum Android version	Android 8.0 (API 26)
Compatibility	Screen size support	4.5" – 6.8"
Scalability	Customers / transactions supported without degradation	1000+ / 50k+
Localisation	Language support (v1.0)	English + Kannada
Battery	Background battery consumption	Negligible (no background sync)

7. Feature Prioritisation (MoSCoW)

Features are categorised using the MoSCoW framework aligned to the v1.0 release.

■ Must Have (MVP — Non-Negotiable)

ID	Feature
M1	Total Outstanding Dues displayed prominently on the Home Screen.
M2	Add Udari (credit) in a maximum 2-step process.
M3	Record customer payments against existing dues.
M4	Customer list searchable by name in real time.
M5	Per-customer transaction history (chronological ledger).
M6	Daily Summary: Total Sold + Dues Pending for today.
M7	Large numeric keypad UI for fast, error-free amount entry.
M8	Offline-first operation — no internet required for core functions.
M9	Room DB persistent local storage with zero data loss.
M10	ViewModel-based live balance calculation.

■ Should Have (Important — High Priority)

ID	Feature
S1	WhatsApp reminder with pre-filled message (vendor name, amount, customer name).
S2	Sort customers by outstanding balance (highest first).

S3	Undo last transaction (within 30 seconds of entry).
S4	Optional note / item tag per transaction (e.g., 'Tomatoes ■80').
S5	Top Debtors widget on Daily Summary screen.
S6	App lock via PIN / biometric (for privacy at market stall).
S7	Date-wise historical summary (past market days).
S8	Kannada language interface option.

■ Could Have (Good to Have — Enhance UX)

ID	Feature
C1	GenAI Smart Insight: auto-flag customers with 2+ market cycles unpaid.
C2	GenAI Plain-Language Summary: 'You are owed 18% more than last week'.
C3	Dark mode / high-contrast mode for outdoor sunlight visibility.
C4	Export ledger as PDF or CSV for sharing / printing.
C5	Customer photo for quick visual identification at the stall.
C6	Quick-dial or SMS fallback if customer has no WhatsApp.
C7	Recurring credit templates (e.g., 'Same as last week for Ramu: ■150').
C8	Animated onboarding tutorial for first-time users.

■ Won't Have (Out of Scope for v1.0)

- ✗ Cloud synchronisation or multi-device support.
- ✗ Inventory management or stock control.
- ✗ Payment gateway / UPI integration.
- ✗ GST invoicing or formal accounting.
- ✗ Web dashboard or desktop version.

8. Technical Implementation

Layer	Technology	Purpose
Language	Kotlin	Primary development language
Architecture	MVVM + Clean Architecture	Separation of concerns; testability
Database	Room DB (SQLite)	Persistent local transaction storage
Async	Kotlin Coroutines + Flow	Non-blocking DB operations; live UI updates

DI	Hilt (Dagger)	Dependency injection for ViewModel / Repo
UI Framework	Jetpack Compose	Declarative UI; fast iteration
Navigation	Compose Navigation	Type-safe screen routing
State Mgmt	ViewModel + StateFlow	Reactive balance calculation; lifecycle-safe
GenAI	Gemini Nano (on-device) / Gemini API	Smart insights, plain-language summaries
WhatsApp	Android Intent (ACTION_SEND)	Open WhatsApp with pre-filled reminder message
Security	EncryptedSharedPreferences + PIN	App lock; secure preference storage
Testing	JUnit 4 + Espresso + Truth	Unit tests for ViewModel; UI tests for critical flows

9. Success Criteria for Student Developers

The following criteria define the minimum bar for a successful student submission. All criteria must pass for the project to be considered complete.

#	Success Criterion	Verification Method	Priority
1	Total Outstanding Dues is visible on the Home Screen without any navigation.	Visual inspection / automated UI test	Critical
2	Ledger is searchable by customer name; results appear in real time.	Search 'Ra' — expect 'Ramesh', 'Raju' etc. within 300 ms	Critical
3	Adding a new Udari transaction is completed in exactly 2 steps (customer selection + amount entry).	User test with a new evaluator; count taps	Critical
4	Net balance per customer updates immediately after any transaction.	Add ■500 Udari; verify balance increases by ■500 instantly	High
5	Daily Summary shows correct totals matching manual calculation.	Insert 5 known transactions; verify summary arithmetic	High
6	WhatsApp reminder opens with pre-filled message containing customer name and amount.	Tap reminder icon; inspect pre-filled WhatsApp text	Medium
7	App works fully offline — no crash or data loss without internet.	Enable airplane mode; perform all core operations	High
8	Data persists across app restarts (Room DB).	Add records; force-kill app; reopen — data intact	Critical

10. Impact Goals & Digital India Alignment

Impact Area	Goal	How Namma-Santhe Ledger Delivers
Financial Inclusion	Bring digital bookkeeping to unorganised retail	Replaces paper diary; no accounting literacy required
Micro-Business Stability	Reduce bad debts for the smallest entrepreneurs	Automated reminders; visibility into aged dues
Digital India	Move Santhe economy from paper to pixels	Offline-first Android app accessible on any Rs 6,000 phone
Women Empowerment	Support women vendors (majority of Santhe sellers)	Simple UI, voice-future, regional language ready
GenAI for Bharat	Demonstrate AI value in grassroots commerce	Gemini-powered insights in plain Kannada / English

■ Namma Santhe. Namma Ledger. Namma Future.

Every rupee tracked is a micro-business protected. Namma-Santhe Ledger turns a ■6,000 Android phone into the most powerful bookkeeping tool a rural vendor has ever had.